CONNECTED EMBEDDED SYSTEM

# ExitLight Health Checker

## ESP32 cellular IoT retrofit for automated emergency-light health testing

I designed and developed a retrofit controller that adds remote test control, light-level sensing and cloud connectivity to an existing emergency exit light. The prototype can interrupt mains power, allow the fitting to operate from its own battery, measure illumination and return the result remotely.

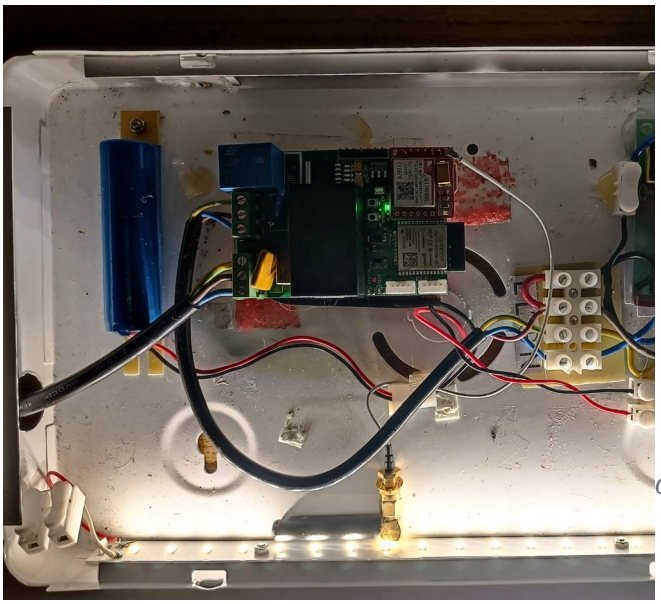


*Working controller prototype; €1 coin shown for scale.*

| | |
|---|---|
| **ESP32-WROOM**<br>Embedded controller | **SIM800L**<br>Cellular / GPRS |
| **BH1750**<br>I²C light sensing | **AWS IoT + MQTT**<br>Remote connectivity |
| **230 VAC relay**<br>Automated test control | **External I²C / GPIO**<br>Expandable sensing |

## Working retrofit prototype

The controller was physically integrated into an OEM emergency-light fitting, demonstrating the complete retrofit concept rather than only a schematic or bench circuit.



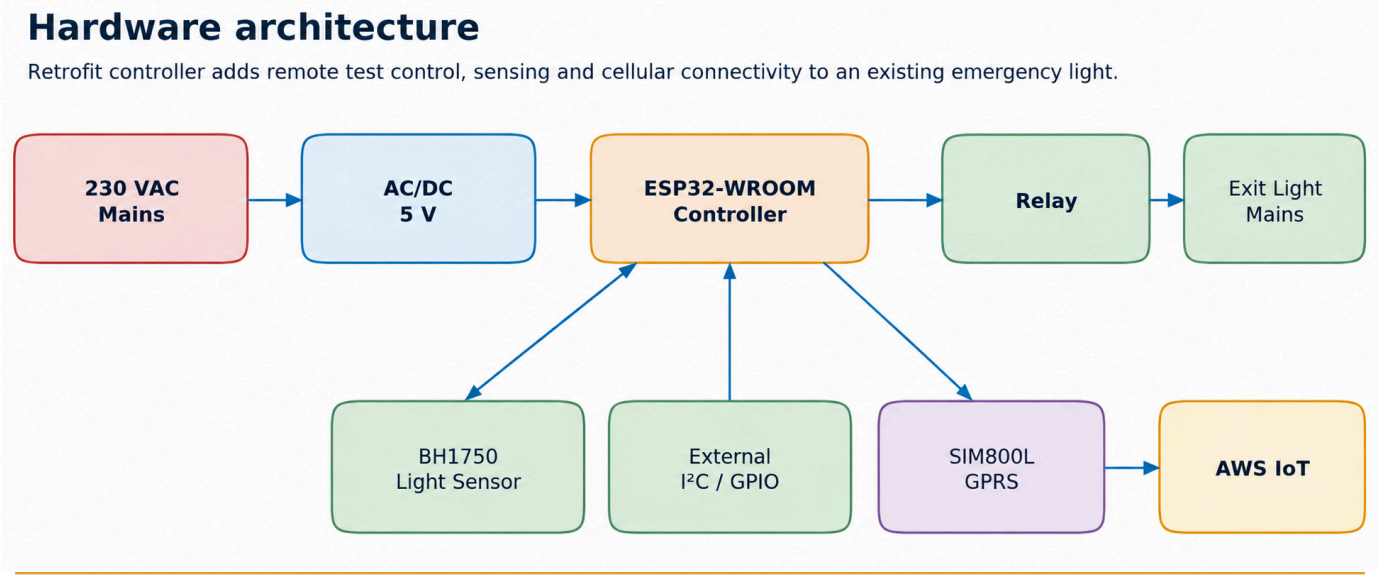
*Prototype installed alongside the original emergency-light electronics.*



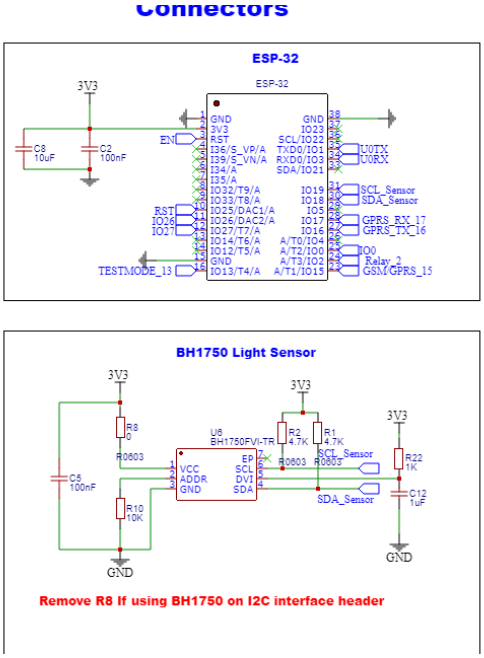
*OEM fitting operating in emergency-light mode.*

# Hardware architecture

The design combines mains-side control, dedicated power rails, an ESP32 controller, local light sensing and cellular communication. The external interface leaves room for additional sensors or peripherals.



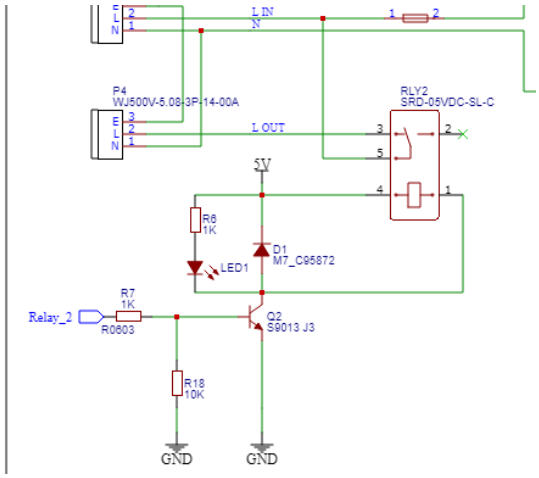

## Controller & sensing




*ESP32 controller and BH1750 I²C light-sensor section from the original schematic.*

- ESP32-WROOM provides the main control and communication functions.
- BH1750 measures light output during the battery test.
- I²C / GPIO plus UART support expansion and programming.

## Power, relay & cellular



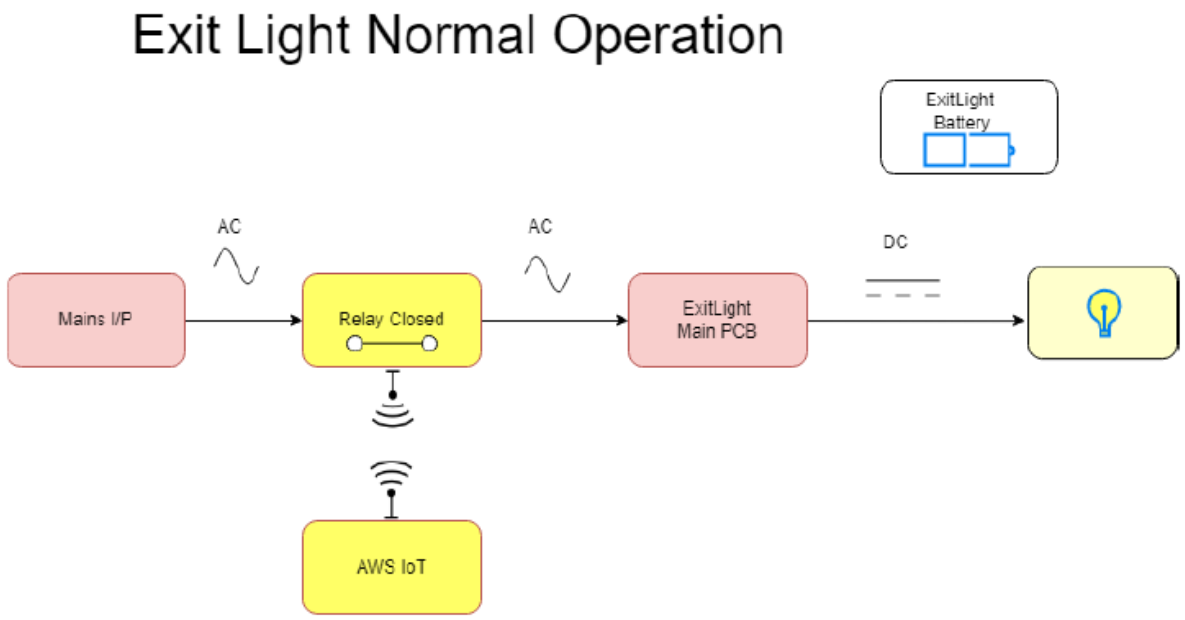
*Mains relay / AC-DC conversion and supporting power circuitry.*

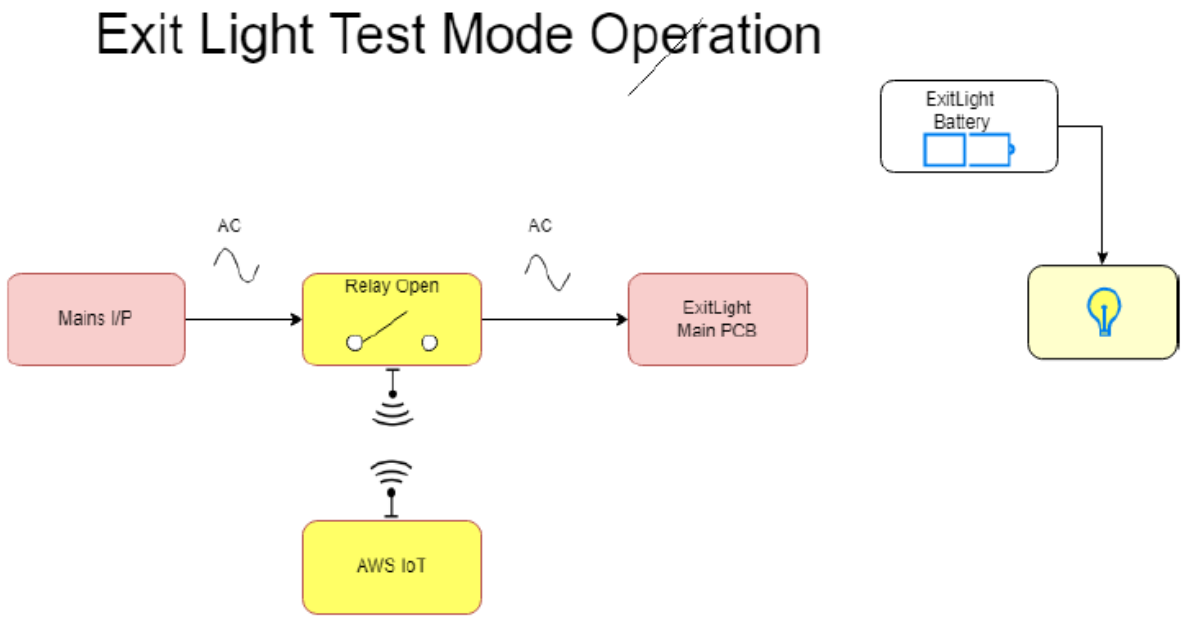
- AC/DC conversion generates a 5 V rail from the mains input.
- Separate regulation supports the ESP32 and SIM800L rails.
- The relay disconnects the OEM fitting; SIM800L provides GPRS connectivity.

**Design intent:** add monitoring and automated test capability without replacing the existing emergency-light electronics.

# Normal Operation & Automated Test

Remote relay control enables a battery-backed emergency-light health check.




## Operating sequence

Normal: the relay remains closed, supplying mains power to the existing exit-light PCB.

Test: a remote command opens the relay, removing mains power from the fitting.

The emergency-light battery then powers the lamp; illumination can be checked by the BH1750.

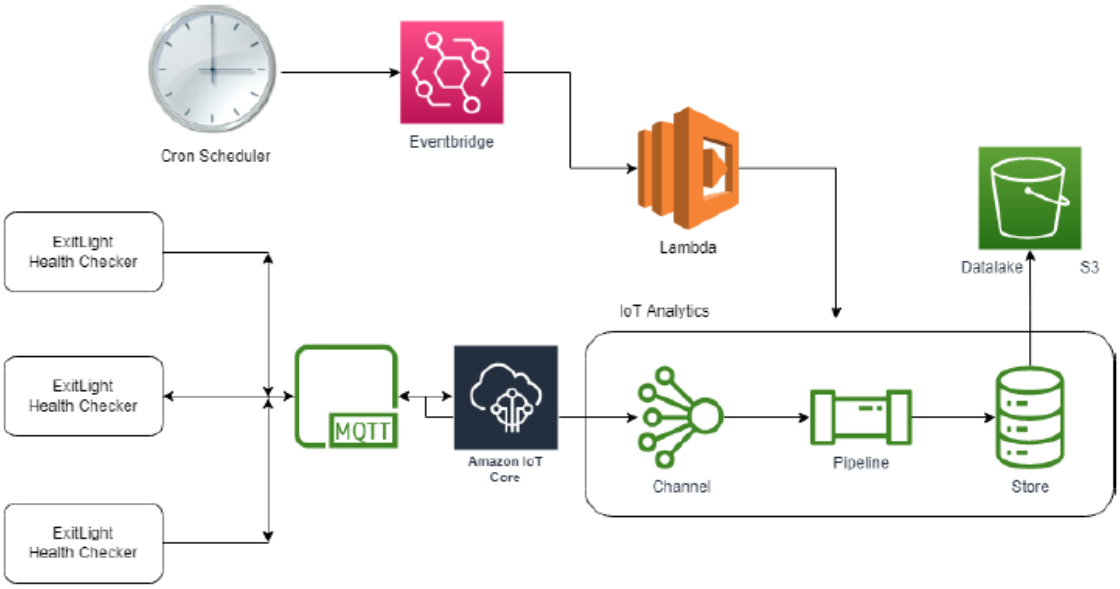
The relay is subsequently returned to its normal state to restore mains operation.

# AWS Cloud Services & Remote Monitoring

Scheduling, cellular communications and cloud-side handling for distributed fittings.

## AWS Cloud Services

## Flowchart




### End-to-end cloud workflow

1. A scheduled event initiates the test workflow through EventBridge and Lambda.
2. AWS IoT Core exchanges MQTT messages with the cellular ExitLight controllers.
3. The controller performs the physical test and returns a light-level measurement.
4. IoT Analytics channels and pipelines support processing and storage of the data.
5. The architecture shows an S3 data-lake destination for retained test information.

### Engineering emphasis

Combines a real retrofit installation with remote control, sensor feedback and an AWS-connected monitoring architecture. Diagram reflects the original project design.